

## I. SUPPLEMENTAL MATERIAL

TABLE I: The Coefficient Matrices of Fault Detection

|             |                                                                                                                                                                                          |             |                                                                                                                                                                                          |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\tilde{Q}$ | $\begin{bmatrix} \tilde{Q}_{11} & \tilde{Q}_{12} & \tilde{Q}_{13} \\ * & \tilde{Q}_{22} & \tilde{Q}_{23} \\ * & * & \tilde{Q}_{33} \end{bmatrix}$                                        | $\tilde{X}$ | $\begin{bmatrix} \tilde{X}_{11} & \tilde{X}_{12} & \tilde{X}_{13} \\ * & \tilde{X}_{22} & \tilde{X}_{23} \\ * & * & \tilde{X}_{33} \end{bmatrix}$                                        |
| $\tilde{S}$ | $\begin{bmatrix} \tilde{S}_{11} & \tilde{S}_{12} & \tilde{S}_{13} \\ \tilde{S}_{21} & \tilde{S}_{22} & \tilde{S}_{23} \\ \tilde{S}_{31} & \tilde{S}_{32} & \tilde{S}_{33} \end{bmatrix}$ | $\tilde{Y}$ | $\begin{bmatrix} \tilde{Y}_{11} & \tilde{Y}_{12} & \tilde{Y}_{13} \\ \tilde{Y}_{21} & \tilde{Y}_{22} & \tilde{Y}_{23} \\ \tilde{Y}_{31} & \tilde{Y}_{32} & \tilde{Y}_{33} \end{bmatrix}$ |
| $\tilde{R}$ | $\begin{bmatrix} \tilde{R}_{11} & \tilde{R}_{12} & \tilde{R}_{13} \\ * & \tilde{R}_{22} & \tilde{R}_{23} \\ * & * & \tilde{R}_{33} \end{bmatrix}$                                        | $\tilde{Z}$ | $\begin{bmatrix} \tilde{Z}_{11} & \tilde{Z}_{12} & \tilde{Z}_{13} \\ * & \tilde{Z}_{22} & \tilde{Z}_{23} \\ * & * & \tilde{Z}_{33} \end{bmatrix}$                                        |

TABLE II: The Coefficient Matrix  $\tilde{Q}$  of Fault Detection in TABLE I

| Coefficient Matrices |                  |       | Values |       |       |       |       |       |       |       |       |                  |                  |                  |
|----------------------|------------------|-------|--------|-------|-------|-------|-------|-------|-------|-------|-------|------------------|------------------|------------------|
| $\tilde{Q}$          | $\tilde{Q}_{11}$ |       | -1.77  | -3.49 | -3.42 | -3.35 | 3.29  | -3.22 | -3.16 | -3.09 | -3.03 | -2.96            | $\times 10^{-5}$ |                  |
|                      |                  |       | -3.49  | -1.71 | -3.35 | -3.28 | -3.22 | -3.15 | -3.08 | -3.01 | -2.95 | -2.88            |                  |                  |
|                      |                  |       | -3.42  | -3.35 | -1.64 | -3.21 | -3.14 | -3.07 | -3.01 | -2.94 | -2.87 | -2.80            |                  |                  |
|                      |                  |       | -3.35  | -3.28 | -3.21 | -1.57 | -3.07 | -3.00 | -2.93 | -2.86 | -2.79 | -2.72            |                  |                  |
|                      |                  |       | -3.29  | -3.22 | -3.14 | -3.07 | -1.50 | -2.93 | -2.86 | -2.79 | -2.71 | -2.64            |                  |                  |
|                      |                  |       | -3.22  | -3.15 | -3.07 | -3.00 | -2.93 | -1.43 | -2.78 | -2.71 | -2.64 | -2.57            |                  |                  |
|                      |                  |       | -3.16  | -3.08 | -3.01 | -2.93 | -2.86 | -2.78 | -1.35 | -2.63 | -2.56 | -2.49            |                  |                  |
|                      |                  |       | -3.09  | -3.01 | -2.94 | -2.86 | -2.79 | -2.71 | -2.63 | -1.28 | -2.48 | -2.41            |                  |                  |
|                      |                  |       | -3.03  | -2.95 | -2.87 | -2.79 | -2.71 | -2.64 | -2.56 | -2.48 | -1.20 | -2.33            |                  |                  |
|                      |                  | -2.96 | -2.88  | -2.80 | -2.72 | -2.64 | -2.57 | -2.49 | -2.41 | -2.33 | -1.13 |                  |                  |                  |
|                      | $\tilde{Q}_{12}$ |       | -2.90  | -2.83 | -2.77 | -2.71 | -2.64 | -2.58 | -2.52 | -2.46 | -2.39 | -2.33            | $\times 10^{-5}$ |                  |
|                      |                  |       | -2.82  | -2.75 | -2.69 | -2.62 | -2.56 | -2.49 | -2.42 | -2.36 | -2.30 | -2.24            |                  |                  |
|                      |                  |       | -2.74  | -2.67 | -2.60 | -2.54 | -2.47 | -2.40 | -2.34 | -2.27 | -2.21 | -2.14            |                  |                  |
|                      |                  |       | -2.65  | -2.59 | -2.52 | -2.45 | -2.38 | -2.32 | -2.25 | -2.18 | -2.12 | -2.05            |                  |                  |
|                      |                  |       | -2.57  | -2.50 | -2.43 | -2.36 | -2.30 | -2.22 | -2.16 | -2.09 | -2.02 | -1.96            |                  |                  |
|                      |                  |       | -2.49  | -2.42 | -2.35 | -2.28 | -2.21 | -2.14 | -2.07 | -2.01 | -1.93 | -1.86            |                  |                  |
|                      |                  |       | -2.41  | -2.34 | -2.27 | -2.20 | -2.13 | -2.05 | -1.98 | -1.91 | -1.84 | -1.77            |                  |                  |
|                      |                  |       | -2.33  | -2.26 | -2.19 | -2.11 | -2.04 | -1.97 | -1.90 | -1.82 | -1.75 | -1.68            |                  |                  |
|                      |                  |       | -2.25  | -2.18 | -2.10 | -2.03 | -1.95 | -1.88 | -1.81 | -1.73 | -1.66 | -1.59            |                  |                  |
|                      |                  | -2.18 | -2.10  | -2.02 | -1.95 | -1.87 | -1.79 | -1.72 | -1.64 | -1.57 | -1.50 |                  |                  |                  |
|                      | $\tilde{Q}_{13}$ |       | -2.27  | -2.21 | -2.15 | -2.09 | -2.03 | -1.97 | -1.91 | -1.85 | -1.80 | -1.74            | -1.68            | $\times 10^{-5}$ |
|                      |                  | -2.18 | -2.11  | -2.05 | -1.99 | -1.93 | -1.87 | -1.81 | -1.75 | -1.69 | -1.63 | -1.57            |                  |                  |
|                      |                  | -2.08 | -2.02  | -1.95 | -1.89 | -1.83 | -1.77 | -1.70 | -1.64 | -1.58 | -1.52 | -1.46            |                  |                  |
|                      |                  | -1.99 | -1.92  | -1.86 | -1.79 | -1.73 | -1.66 | -1.60 | -1.54 | -1.47 | -1.41 | -1.35            |                  |                  |
|                      |                  | -1.89 | -1.82  | -1.76 | -1.69 | -1.63 | -1.56 | -1.50 | -1.43 | -1.37 | -1.31 | -1.24            |                  |                  |
|                      |                  | -1.80 | -1.73  | -1.66 | -1.59 | -1.53 | -1.46 | -1.40 | -1.33 | -1.27 | -1.20 | -1.14            |                  |                  |
|                      |                  | -1.70 | -1.63  | -1.56 | -1.50 | -1.43 | -1.36 | -1.29 | -1.23 | -1.16 | -1.09 | -1.03            |                  |                  |
|                      |                  | -1.61 | -1.54  | -1.47 | -1.40 | -1.33 | -1.26 | -1.19 | -1.12 | -1.06 | -0.99 | -0.92            |                  |                  |
|                      |                  | -1.52 | -1.44  | -1.37 | -1.30 | -1.23 | -1.16 | -1.09 | -1.02 | -0.95 | -0.88 | -0.81            |                  |                  |
|                      | -1.42            | -1.35 | -1.28  | -1.20 | -1.13 | -1.06 | -0.99 | -0.92 | -0.85 | -0.78 | -0.71 |                  |                  |                  |
| $\tilde{Q}_{22}$     |                  | -1.05 | -2.02  | -1.94 | -1.86 | -1.79 | -1.71 | -1.63 | -1.56 | -1.48 | -1.40 | $\times 10^{-5}$ |                  |                  |
|                      |                  | -2.02 | -0.97  | -1.86 | -1.78 | -1.70 | -1.62 | -1.55 | -1.47 | -1.39 | -1.31 |                  |                  |                  |
|                      |                  | -1.94 | -1.86  | -0.89 | -1.70 | -1.62 | -1.54 | -1.46 | -1.38 | -1.30 | -1.22 |                  |                  |                  |
|                      |                  | -1.86 | -1.78  | -1.70 | -0.81 | -1.53 | -1.45 | -1.37 | -1.29 | -1.21 | -1.13 |                  |                  |                  |
|                      |                  | -1.79 | -1.70  | -1.62 | -1.53 | -0.73 | -1.37 | -1.29 | -1.21 | -1.12 | -1.04 |                  |                  |                  |
|                      |                  | -1.71 | -1.62  | -1.54 | -1.45 | -1.37 | -0.64 | -1.20 | -1.12 | -1.04 | -0.95 |                  |                  |                  |
|                      |                  | -1.63 | -1.55  | -1.46 | -1.37 | -1.29 | -1.20 | -0.56 | -1.03 | -0.95 | -0.86 |                  |                  |                  |
|                      |                  | -1.56 | -1.47  | -1.38 | -1.29 | -1.21 | -1.12 | -1.03 | -0.47 | -0.86 | -0.78 |                  |                  |                  |
|                      |                  | -1.48 | -1.39  | -1.30 | -1.21 | -1.12 | -1.04 | -0.95 | -0.86 | -0.39 | -0.69 |                  |                  |                  |
|                      | -1.40            | -1.31 | -1.22  | -1.13 | -1.04 | -0.95 | -0.86 | -0.78 | -0.69 | -0.30 |       |                  |                  |                  |
| $\tilde{Q}_{23}$     |                  | -1.33 | -1.25  | -1.18 | -1.11 | -1.03 | -0.96 | -0.89 | -0.81 | -0.74 | -0.67 | -0.60            | $\times 10^{-5}$ |                  |
|                      |                  | -1.24 | -1.16  | -1.08 | -1.01 | -0.93 | -0.86 | -0.79 | -0.71 | -0.64 | -0.57 | -0.49            |                  |                  |
|                      |                  | -1.14 | -1.07  | -0.99 | -0.91 | -0.84 | -0.76 | -0.69 | -0.61 | -0.54 | -0.46 | -0.39            |                  |                  |
|                      |                  | -1.05 | -0.97  | -0.90 | -0.82 | -0.74 | -0.66 | -0.59 | -0.51 | -0.43 | -0.36 | -0.28            |                  |                  |
|                      |                  | -0.96 | -0.88  | -0.80 | -0.72 | -0.64 | -0.56 | -0.49 | -0.41 | -0.33 | -0.25 | -0.18            |                  |                  |
|                      |                  | -0.87 | -0.79  | -0.71 | -0.63 | -0.55 | -0.47 | -0.39 | -0.31 | -0.23 | -0.15 | -0.07            |                  |                  |
|                      |                  | -0.78 | -0.70  | -0.61 | -0.53 | -0.45 | -0.37 | -0.29 | -0.21 | -0.13 | -0.05 | 0.03             |                  |                  |
|                      |                  | -0.69 | -0.61  | -0.52 | -0.44 | -0.36 | -0.27 | -0.19 | -0.11 | -0.03 | 0.05  | 0.13             |                  |                  |
|                      |                  | -0.60 | -0.52  | -0.43 | -0.34 | -0.26 | -0.18 | -0.09 | -0.01 | 0.07  | 0.15  | 0.24             |                  |                  |
|                      | -0.51            | -0.42 | -0.34  | -0.25 | -0.17 | -0.08 | 0.00  | 0.09  | 0.17  | 0.26  | 0.34  |                  |                  |                  |
| $\tilde{Q}_{33}$     |                  | -2.12 | -3.34  | -2.46 | -1.59 | -0.71 | 0.15  | 1.02  | 1.88  | 2.73  | 3.58  | 4.43             | $\times 10^{-5}$ |                  |
|                      |                  | -3.34 | -1.23  | -1.55 | -0.66 | 0.22  | 1.11  | 1.98  | 2.86  | 3.73  | 4.59  | 5.45             |                  |                  |
|                      |                  | -2.46 | -1.55  | -0.33 | 0.26  | 1.16  | 2.05  | 2.95  | 3.83  | 4.72  | 5.59  | 6.46             |                  |                  |
|                      |                  | -1.59 | -0.66  | 0.26  | 0.58  | 2.09  | 3.00  | 3.91  | 4.81  | 5.70  | 6.59  | 7.47             |                  |                  |
|                      |                  | -0.71 | 0.22   | 1.16  | 2.09  | 1.50  | 3.94  | 4.86  | 5.77  | 6.68  | 7.58  | 8.48             |                  |                  |
|                      |                  | 0.15  | 1.11   | 2.05  | 3.00  | 3.94  | 2.43  | 5.81  | 6.73  | 7.65  | 8.57  | 9.49             |                  |                  |
|                      |                  | 1.02  | 1.98   | 2.95  | 3.91  | 4.86  | 5.81  | 3.37  | 7.69  | 8.63  | 9.56  | 10.49            |                  |                  |
|                      |                  | 1.88  | 2.86   | 3.83  | 4.81  | 5.77  | 6.73  | 7.69  | 4.32  | 9.60  | 10.55 | 11.49            |                  |                  |
|                      |                  | 2.73  | 3.73   | 4.72  | 5.70  | 6.68  | 7.65  | 8.63  | 9.60  | 5.28  | 11.53 | 12.49            |                  |                  |
|                      |                  | 3.58  | 4.59   | 5.59  | 6.59  | 7.58  | 8.57  | 9.56  | 10.55 | 11.53 | 6.24  | 13.48            |                  |                  |
|                      |                  | 4.43  | 5.45   | 6.46  | 7.47  | 8.48  | 9.49  | 10.49 | 11.49 | 12.49 | 13.48 | 7.22             |                  |                  |

TABLE III: The Coefficient Matrix  $\tilde{S}$  of Fault Detection in TABLE I

| Coefficient Matrices |        | Values           |       |        |        |        |        |        |        |        |        |                  |        |                  |                  |
|----------------------|--------|------------------|-------|--------|--------|--------|--------|--------|--------|--------|--------|------------------|--------|------------------|------------------|
| $\tilde{S}_{11}$     |        | -5.02            | -8.09 | -8.51  | -8.56  | -21.24 | -24.39 | -24.71 | -24.77 | -23.74 | -21.55 | $\times 10^{-4}$ |        |                  |                  |
|                      |        | -4.12            | -7.23 | -7.63  | -7.71  | -20.45 | -23.64 | -23.95 | -24.04 | -22.98 | -20.79 |                  |        |                  |                  |
|                      |        | -3.23            | -6.36 | -6.76  | -6.87  | -19.67 | -22.89 | -23.19 | -23.32 | -22.23 | -20.04 |                  |        |                  |                  |
|                      |        | -2.34            | -5.50 | -5.90  | -6.03  | -18.89 | -22.14 | -22.44 | -22.59 | -21.48 | -19.28 |                  |        |                  |                  |
|                      |        | -1.46            | -4.65 | -5.04  | -5.20  | -18.11 | -21.39 | -21.69 | -21.87 | -20.73 | -18.53 |                  |        |                  |                  |
|                      |        | -0.54            | -3.80 | -4.18  | -4.37  | -17.34 | -20.65 | -20.95 | -21.16 | -19.99 | -17.78 |                  |        |                  |                  |
|                      |        | 0.37             | -2.95 | -3.33  | -3.54  | -16.57 | -19.92 | -20.21 | -20.44 | -19.24 | -17.05 |                  |        |                  |                  |
|                      |        | 1.29             | -2.12 | -2.49  | -2.72  | -15.80 | -19.18 | -19.47 | -19.73 | -18.51 | -16.31 |                  |        |                  |                  |
|                      |        | 2.20             | -1.27 | -1.64  | -1.91  | -15.04 | -18.45 | -18.73 | -19.03 | -17.77 | -15.57 |                  |        |                  |                  |
|                      |        | 3.12             | -0.39 | -0.75  | -1.07  | -14.28 | -17.73 | -18.00 | -18.32 | -17.04 | -14.84 |                  |        |                  |                  |
|                      |        | $\tilde{S}_{12}$ |       | -21.24 | -18.36 | -17.63 | -16.11 | -15.99 | -13.29 | -12.70 | -11.37 |                  | -11.44 | -8.92            | $\times 10^{-4}$ |
|                      |        |                  |       | -20.47 | -17.60 | -16.84 | -15.32 | -15.19 | -12.50 | -11.88 | -10.55 |                  | -10.61 | -8.09            |                  |
| -19.70               | -16.84 |                  |       | -16.05 | -14.54 | -14.39 | -11.71 | -11.06 | -9.73  | -9.78  | -7.27  |                  |        |                  |                  |
| -18.93               | -16.08 |                  |       | -15.27 | -13.75 | -13.60 | -10.92 | -10.24 | -8.92  | -8.95  | -6.45  |                  |        |                  |                  |
| -18.17               | -15.32 |                  |       | -14.49 | -12.97 | -12.80 | -10.13 | -9.43  | -8.11  | -8.13  | -5.63  |                  |        |                  |                  |
| -17.41               | -14.57 |                  |       | -13.71 | -12.20 | -12.02 | -9.35  | -8.62  | -7.30  | -7.31  | -4.82  |                  |        |                  |                  |
| -16.66               | -13.82 |                  |       | -12.93 | -11.43 | -11.23 | -8.57  | -7.82  | -6.50  | -6.50  | -4.02  |                  |        |                  |                  |
| -15.91               | -13.08 |                  |       | -12.16 | -10.66 | -10.45 | -7.80  | -7.02  | -5.71  | -5.69  | -3.21  |                  |        |                  |                  |
| -15.16               | -12.34 |                  |       | -11.40 | -9.90  | -9.67  | -7.03  | -6.22  | -4.91  | -4.88  | -2.41  |                  |        |                  |                  |
| -14.41               | -11.60 |                  |       | -10.64 | -9.14  | -8.90  | -6.26  | -5.43  | -4.13  | -4.08  | -1.62  |                  |        |                  |                  |
| $\tilde{S}_{13}$     |        |                  |       | -8.47  | -7.32  | -7.59  | -5.24  | -4.94  | -3.97  | -4.42  | -2.25  | -2.10            | -1.31  | $\times 10^{-4}$ |                  |
|                      |        |                  |       | -7.62  | -6.47  | -6.72  | -4.38  | -4.05  | -3.09  | -3.52  | -1.35  | -1.18            | -0.40  |                  |                  |
|                      |        | -6.77            | -5.62 | -5.86  | -3.52  | -3.17  | -2.21  | -2.63  | -0.47  | -0.26  | 0.52   |                  |        |                  |                  |
|                      |        | -5.92            | -4.78 | -5.00  | -2.67  | -2.29  | -1.33  | -1.74  | 0.42   | 0.65   | 1.43   |                  |        |                  |                  |
|                      |        | -5.08            | -3.94 | -4.15  | -1.83  | -1.41  | -0.46  | -0.85  | 1.30   | 1.57   | 2.33   |                  |        |                  |                  |
|                      |        | -4.24            | -3.10 | -3.30  | -0.98  | -0.54  | 0.41   | 0.03   | 2.17   | 2.45   | 3.21   |                  |        |                  |                  |
|                      |        | -3.40            | -2.27 | -2.45  | -0.14  | 0.32   | 1.27   | 0.91   | 3.04   | 3.34   | 4.10   |                  |        |                  |                  |
|                      |        | -2.57            | -1.45 | -1.61  | 0.69   | 1.19   | 2.12   | 1.78   | 3.91   | 4.22   | 4.99   |                  |        |                  |                  |
|                      |        | -1.75            | -0.62 | -0.78  | 1.52   | 2.04   | 2.97   | 2.65   | 4.79   | 5.10   | 5.87   |                  |        |                  |                  |
|                      |        | -0.93            | 0.19  | 0.06   | 2.36   | 2.88   | 3.83   | 3.52   | 5.66   | 5.98   | 6.76   |                  |        |                  |                  |
|                      |        | $\tilde{S}_{21}$ |       | 0.40   | 0.05   | 0.01   | -0.02  | -13.53 | -17.01 | -17.27 | -17.62 | -16.32           | -14.11 |                  | $\times 10^{-4}$ |
|                      |        |                  |       | 0.49   | 0.13   | 0.10   | 0.06   | -12.78 | -16.29 | -16.55 | -16.92 | -15.59           | -13.39 |                  |                  |
| 0.59                 | 0.22   |                  |       | 0.19   | 0.15   | -12.03 | -15.58 | -15.83 | -16.23 | -14.87 | -12.67 |                  |        |                  |                  |
| 0.68                 | 0.31   |                  |       | 0.28   | 0.24   | -11.29 | -14.87 | -15.11 | -15.54 | -14.16 | -11.96 |                  |        |                  |                  |
| 0.77                 | 0.40   |                  |       | 0.37   | 0.32   | -10.55 | -14.16 | -14.40 | -14.86 | -13.45 | -11.25 |                  |        |                  |                  |
| 0.86                 | 0.48   |                  |       | 0.46   | 0.41   | -9.82  | -13.46 | -13.69 | -14.17 | -12.74 | -10.54 |                  |        |                  |                  |
| 0.95                 | 0.57   |                  |       | 0.54   | 0.49   | -9.09  | -12.76 | -12.99 | -13.49 | -12.04 | -9.84  |                  |        |                  |                  |
| 1.04                 | 0.65   |                  |       | 0.63   | 0.58   | -8.37  | -12.07 | -12.29 | -12.82 | -11.34 | -9.14  |                  |        |                  |                  |
| 1.13                 | 0.74   |                  |       | 0.72   | 0.66   | -7.65  | -11.38 | -11.59 | -12.15 | -10.65 | -8.45  |                  |        |                  |                  |
| 1.22                 | 0.83   |                  |       | 0.81   | 0.74   | -6.93  | -10.70 | -10.90 | -11.48 | -9.96  | -7.76  |                  |        |                  |                  |
| $\tilde{S}_{22}$     |        |                  |       | -13.67 | -10.87 | -9.88  | -8.38  | -8.13  | -5.50  | -4.65  | -3.34  | -3.28            | -0.82  | $\times 10^{-4}$ |                  |
|                      |        |                  |       | -12.94 | -10.14 | -9.12  | -7.63  | -7.37  | -4.74  | -3.86  | -2.56  | -2.49            | -0.04  |                  |                  |
|                      |        | -12.21           | -9.41 | -8.37  | -6.88  | -6.61  | -3.99  | -3.08  | -1.79  | -1.70  | 0.75   |                  |        |                  |                  |
|                      |        | -11.48           | -8.69 | -7.63  | -6.14  | -5.85  | -3.24  | -2.31  | -1.02  | -0.91  | 1.55   |                  |        |                  |                  |
|                      |        | -10.75           | -7.97 | -6.89  | -5.40  | -5.10  | -2.49  | -1.54  | -0.26  | -0.13  | 2.34   |                  |        |                  |                  |
|                      |        | -10.03           | -7.26 | -6.15  | -4.67  | -4.35  | -1.75  | -0.78  | 0.51   | 0.66   | 3.14   |                  |        |                  |                  |
|                      |        | -9.32            | -6.55 | -5.42  | -3.94  | -3.61  | -1.01  | -0.02  | 1.29   | 1.45   | 3.93   |                  |        |                  |                  |
|                      |        | -8.61            | -5.84 | -4.69  | -3.22  | -2.87  | -0.28  | 0.75   | 2.06   | 2.24   | 4.72   |                  |        |                  |                  |
|                      |        | -7.90            | -5.14 | -3.97  | -2.50  | -2.14  | 0.47   | 1.51   | 2.83   | 3.03   | 5.51   |                  |        |                  |                  |
|                      |        | -7.20            | -4.45 | -3.25  | -1.79  | -1.41  | 1.22   | 2.28   | 3.60   | 3.82   | 6.29   |                  |        |                  |                  |
|                      |        | $\tilde{S}_{23}$ |       | -0.11  | 1.01   | 0.89   | 3.20   | 3.73   | 4.68   | 4.39   | 6.52   | 6.86             | 7.64   |                  | $\times 10^{-4}$ |
|                      |        |                  |       | 0.70   | 1.82   | 1.72   | 4.03   | 4.57   | 5.53   | 5.26   | 7.39   | 7.74             | 8.53   |                  |                  |
| 1.51                 | 2.64   |                  |       | 2.55   | 4.86   | 5.42   | 6.38   | 6.13   | 8.26   | 8.62   | 9.41   |                  |        |                  |                  |
| 2.32                 | 3.45   |                  |       | 3.39   | 5.69   | 6.26   | 7.23   | 7.00   | 9.13   | 9.50   | 10.30  |                  |        |                  |                  |
| 3.13                 | 4.26   |                  |       | 4.22   | 6.52   | 7.10   | 8.08   | 7.87   | 9.99   | 10.37  | 11.18  |                  |        |                  |                  |
| 3.93                 | 5.07   |                  |       | 5.05   | 7.35   | 7.95   | 8.92   | 8.73   | 10.85  | 11.25  | 12.06  |                  |        |                  |                  |
| 4.74                 | 5.88   |                  |       | 5.88   | 8.18   | 8.79   | 9.77   | 9.59   | 11.71  | 12.12  | 12.94  |                  |        |                  |                  |
| 5.54                 | 6.69   |                  |       | 6.71   | 9.00   | 9.62   | 10.61  | 10.46  | 12.57  | 12.99  | 13.82  |                  |        |                  |                  |
| 6.34                 | 7.50   |                  |       | 7.53   | 9.83   | 10.46  | 11.46  | 11.32  | 13.43  | 13.86  | 14.69  |                  |        |                  |                  |
| 7.14                 | 8.31   |                  |       | 8.36   | 10.65  | 11.29  | 12.30  | 12.18  | 14.28  | 14.73  | 15.56  |                  |        |                  |                  |
| $\tilde{S}_{31}$     |        |                  |       | 13.11  | 9.13   | 8.94   | 8.28   | -6.23  | -10.02 | -10.21 | -10.82 | -9.27            | -7.08  | $\times 10^{-4}$ |                  |
|                      |        |                  |       | 14.01  | 9.99   | 9.81   | 9.12   | -5.52  | -9.34  | -9.53  | -10.16 | -8.59            | -6.40  |                  |                  |
|                      |        | 14.91            | 10.85 | 10.68  | 9.96   | -4.82  | -8.67  | -8.85  | -9.51  | -7.92  | -5.73  |                  |        |                  |                  |
|                      |        | 15.80            | 11.70 | 11.54  | 10.80  | -4.13  | -8.01  | -8.18  | -8.86  | -7.24  | -5.06  |                  |        |                  |                  |
|                      |        | 16.69            | 12.55 | 12.41  | 11.63  | -3.44  | -7.35  | -7.51  | -8.21  | -6.58  | -4.40  |                  |        |                  |                  |
|                      |        | 17.58            | 13.40 | 13.27  | 12.46  | -2.75  | -6.69  | -6.85  | -7.57  | -5.92  | -3.74  |                  |        |                  |                  |
|                      |        | 18.46            | 14.25 | 14.13  | 13.29  | -2.07  | -6.04  | -6.19  | -6.93  | -5.26  | -3.09  |                  |        |                  |                  |
|                      |        | 19.35            | 15.09 | 14.98  | 14.11  | -1.37  | -5.40  | -5.55  | -6.30  | -4.61  | -2.44  |                  |        |                  |                  |
|                      |        | 20.23            | 15.93 | 15.84  | 14.93  | -0.66  | -4.76  | -4.89  | -5.68  | -3.96  | -1.79  |                  |        |                  |                  |
|                      |        | 21.10            | 16.77 | 16.69  | 15.75  | 0.05   | -4.13  | -4.24  | -5.05  | -3.32  | -1.13  |                  |        |                  |                  |
|                      |        | 21.98            | 17.61 | 17.54  | 16.57  | 0.76   | -3.50  | -3.60  | -4.43  | -2.69  | -0.45  |                  |        |                  |                  |
|                      |        | $\tilde{S}_{32}$ |       | -6.50  | -3.75  | -2.53  | -1.08  | -0.68  | 1.97   | 3.04   | 4.37   | 4.61             | 7.08   |                  | $\times 10^{-4}$ |
| -5.81                | -3.07  |                  |       | -1.83  | -0.36  | 0.07   | 2.72   | 3.80   | 5.13   | 5.39   | 7.86   |                  |        |                  |                  |
| -5.12                | -2.38  |                  |       | -1.12  | 0.36   | 0.81   | 3.46   | 4.56   | 5.90   | 6.18   | 8.64   |                  |        |                  |                  |
| -4.43                | -1.70  |                  |       | -0.41  | 1.09   | 1.56   | 4.20   | 5.31   | 6.66   | 6.96   | 9.42   |                  |        |                  |                  |
| -3.76                | -1.03  |                  |       | 0.30   | 1.81   | 2.30   | 4.94   | 6.07   | 7.42   | 7.74   | 10.20  |                  |        |                  |                  |
| -3.08                | -0.32  |                  |       | 1.02   | 2.53   | 3.04   | 5.68   | 6.82   | 8.18   | 8.51   | 10.97  |                  |        |                  |                  |
| -2.41                | 0.37   |                  |       | 1.73   | 3.25   | 3.78   | 6.42   | 7.57   | 8.94   | 9.29   | 11.74  |                  |        |                  |                  |
| -1.75                | 1.07   |                  |       | 2.44   | 3.97   | 4.51   | 7.15   | 8.31   | 9.69   | 10.06  | 12.51  |                  |        |                  |                  |
| -1.07                | 1.76   |                  |       | 3.15   | 4.68   | 5.24   | 7.88   | 9.06   | 10.44  | 10.83  | 13.27  |                  |        |                  |                  |
| -0.37                | 2.45   |                  |       | 3.85   | 5.40   | 5.97   | 8.61   | 9.80   | 11.19  | 11.59  | 14.03  |                  |        |                  |                  |
| 0.32                 | 3.14   |                  |       | 4.55   | 6.10   | 6.70   | 9.33   | 10.54  | 11.93  | 12.35  | 14.79  |                  |        |                  |                  |
| $\tilde{S}_{33}$     |        |                  |       | 7.94   | 9.11   | 9.18   | 11.47  | 12.13  | 13.14  | 13.03  | 15.13  | 15.59            | 16.44  | $\times 10^{-4}$ |                  |
|                      |        | 8.74             | 9.91  | 10.00  | 12.28  | 12.96  | 13.97  | 13.89  | 15.98  | 16.46  | 17.31  |                  |        |                  |                  |
|                      |        | 9.53             | 10.72 | 10.82  | 13.10  | 13.79  | 14.81  | 14.74  | 16.83  | 17.32  | 18.17  |                  |        |                  |                  |
|                      |        | 10.32            | 11.51 | 11.64  | 13.91  | 14.61  | 15.64  | 15.59  | 17.67  | 18.18  | 19.04  |                  |        |                  |                  |
|                      |        | 11.11            | 12.31 | 12.45  | 14.72  | 15.43  | 16.47  | 16.43  | 18.52  | 19.03  | 19.90  |                  |        |                  |                  |
|                      |        | 11.90            | 13.10 | 13.26  | 15.53  | 16.26  | 17.30  | 17.28  | 19.36  | 19.89  | 20.76  |                  |        |                  |                  |
|                      |        | 12.68            | 13.89 | 14.07  | 16.33  | 17.07  | 18.12  | 18.12  | 20.19  | 20.74  | 21.62  |                  |        |                  |                  |
|                      |        | 13.46            | 14.68 | 14.87  | 17.13  | 17.89  | 18.94  | 18.95  | 21.02  | 21.58  | 22.47  |                  |        |                  |                  |
|                      |        | 14.24            | 15.47 | 15.67  | 17.93  | 18.70  | 19.76  | 19.79  | 21.85  | 22.43  | 23.32  |                  |        |                  |                  |
|                      |        | 15.02            | 16.25 | 16.47  | 18.72  | 19.51  | 20.58  | 20.62  | 22.68  | 23.27  | 24.17  |                  |        |                  |                  |
|                      |        | 15.79            | 17.03 | 17.27  | 19.52  | 20.32  | 21.39  | 21.45  | 23.50  | 24.11  | 25.01  |                  |        |                  |                  |

TABLE IV: The Coefficient Matrix  $\tilde{R}$  of Fault Detection in TABLE I

| Coefficient Matrices |                  | Values |        |        |        |        |        |        |        |        |        |        |                  |
|----------------------|------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|------------------|
| $\tilde{R}$          | $\tilde{R}_{11}$ |        | -13.42 | 8.25   | 11.01  | 11.45  | -13.17 | 13.77  | 0.98   | 13.44  | -19.20 | 14.29  | $\times 10^{-3}$ |
|                      |                  |        | 8.25   | -15.38 | 9.00   | -34.43 | 13.79  | -8.25  | 13.36  | 11.69  | 10.45  | -19.44 |                  |
|                      |                  |        | 11.01  | 9.00   | -13.77 | 12.46  | 1.31   | 13.32  | -11.98 | 12.07  | 13.42  | 10.95  |                  |
|                      |                  |        | 11.45  | -34.43 | 12.46  | -12.38 | 13.70  | 10.84  | 12.48  | -14.06 | 8.34   | 15.84  |                  |
|                      |                  |        | -13.17 | 13.79  | 1.31   | 13.70  | -18.66 | 2.52   | 14.60  | 1.46   | -0.73  | 2.44   |                  |
|                      |                  |        | 13.77  | -8.25  | 13.32  | 10.84  | 2.52   | -18.30 | 1.16   | 15.01  | -4.47  | -0.35  |                  |
|                      |                  |        | 0.98   | 13.36  | -11.98 | 12.48  | 14.60  | 1.16   | -18.39 | -1.05  | 12.07  | -3.93  |                  |
|                      |                  |        | 13.44  | 11.69  | 12.07  | -14.06 | 1.46   | 15.01  | -1.05  | -18.22 | -7.17  | 10.03  |                  |
|                      |                  |        | -19.20 | 10.45  | 13.42  | 8.34   | -0.73  | -4.47  | 12.07  | -7.17  | -17.24 | -5.55  |                  |
|                      |                  |        | 14.29  | -19.44 | 10.95  | 15.84  | 2.44   | -0.35  | -3.93  | 10.03  | -5.55  | -16.91 |                  |
| $\tilde{R}_{12}$     |                  |        | 8.43   | 14.43  | -18.28 | 14.89  | 12.36  | 14.98  | -9.53  | 15.46  | 14.95  | 15.52  | $\times 10^{-3}$ |
|                      |                  |        | 14.15  | 14.65  | 11.71  | -19.04 | 14.74  | 16.63  | 12.27  | -11.08 | 15.29  | 17.24  |                  |
|                      |                  |        | -19.37 | 12.81  | 15.22  | 11.53  | -18.27 | 13.33  | 15.82  | 12.09  | -9.76  | 13.83  |                  |
|                      |                  |        | 9.78   | -19.37 | 9.67   | 16.49  | 10.90  | -13.89 | 10.81  | 16.33  | 11.98  | -2.17  |                  |
|                      |                  |        | 14.94  | 2.86   | -15.53 | 3.12   | 13.89  | 3.51   | -18.71 | 3.78   | 11.84  | 4.13   |                  |
|                      |                  |        | 2.06   | 13.92  | -2.97  | -15.09 | 2.66   | 11.40  | -2.34  | -19.16 | 3.23   | 8.14   |                  |
|                      |                  |        | -1.52  | -0.15  | 9.70   | -3.41  | -15.44 | 0.44   | 6.43   | -2.87  | -18.72 | 1.02   |                  |
|                      |                  |        | -5.54  | -9.64  | -5.78  | 7.29   | -4.28  | -18.06 | -4.51  | 4.25   | -2.99  | -15.40 |                  |
|                      |                  |        | 7.93   | -6.56  | -17.58 | -5.06  | 4.47   | -6.06  | -16.21 | -4.55  | 1.01   | -5.55  |                  |
|                      |                  |        | -4.27  | 6.59   | -4.93  | -18.61 | -3.84  | 2.75   | -4.52  | -15.37 | -3.40  | -0.98  |                  |
| $\tilde{R}_{13}$     |                  |        | 2.43   | 16.02  | 16.28  | 16.04  | 13.03  | 16.57  | 16.44  | 16.54  | 16.12  | 17.09  | $\times 10^{-3}$ |
|                      |                  |        | 12.79  | 0.34   | 15.82  | 16.57  | 13.28  | 11.63  | 16.31  | 14.79  | 13.73  | 16.10  |                  |
|                      |                  |        | 15.20  | 12.63  | 1.90   | 14.31  | 13.51  | 13.15  | 12.70  | 14.78  | 10.96  | 13.66  |                  |
|                      |                  |        | 11.91  | 15.40  | 13.01  | 10.34  | 12.97  | 13.80  | 14.00  | 16.64  | 13.99  | 11.62  |                  |
|                      |                  |        | -12.91 | 4.42   | 9.02   | 4.74   | -1.59  | 5.04   | 5.78   | 5.33   | 10.14  | 5.65   |                  |
|                      |                  |        | -1.70  | -14.46 | 3.78   | 4.62   | -1.03  | -3.73  | 4.31   | 1.45   | -0.35  | 8.14   |                  |
|                      |                  |        | 2.93   | -2.32  | -13.33 | 1.59   | -0.47  | -1.76  | -2.40  | 2.14   | -4.56  | -1.18  |                  |
|                      |                  |        | -3.22  | 1.17   | -1.68  | -4.85  | -1.90  | -1.78  | -0.35  | 7.84   | -0.55  | -4.75  |                  |
|                      |                  |        | -7.31  | -4.02  | -2.81  | -5.03  | 4.64   | -3.49  | -6.72  | -4.50  | 14.76  | -2.95  |                  |
|                      |                  |        | -4.09  | -5.72  | -2.95  | -4.90  | -3.64  | 6.17   | -2.49  | -8.40  | -3.18  | 15.50  |                  |
| $\tilde{R}_{22}$     |                  |        | -17.10 | -5.96  | 4.01   | -4.51  | -18.40 | -5.60  | 0.45   | -4.14  | -15.94 | -5.24  | $\times 10^{-3}$ |
|                      |                  |        | -5.96  | -14.70 | -6.37  | 2.77   | -5.46  | -16.73 | -5.27  | -0.35  | -4.34  | -7.74  |                  |
|                      |                  |        | 4.01   | -6.37  | -10.63 | -5.63  | -0.11  | -6.61  | -9.91  | -5.27  | -4.29  | -6.25  |                  |
|                      |                  |        | -4.51  | 2.77   | -5.63  | -9.95  | -5.30  | -1.73  | -5.97  | -8.62  | -5.04  | -5.71  |                  |
|                      |                  |        | -18.40 | -5.46  | -0.11  | -5.30  | -10.37 | -6.31  | -5.08  | -5.70  | -9.51  | -6.09  |                  |
|                      |                  |        | -5.60  | -16.73 | -6.61  | -1.73  | -6.31  | -5.38  | -6.69  | -5.64  | -5.96  | 1.69   |                  |
|                      |                  |        | 0.45   | -5.27  | -9.91  | -5.97  | -5.08  | -6.69  | -0.62  | -6.77  | -9.53  | -7.04  |                  |
|                      |                  |        | -4.14  | -0.35  | -5.27  | -8.62  | -5.70  | -5.64  | -6.77  | -0.01  | -6.60  | -9.72  |                  |
|                      |                  |        | -15.94 | -4.34  | -4.29  | -5.04  | -9.51  | -5.96  | -9.53  | -6.60  | -0.47  | -6.90  |                  |
|                      |                  |        | -5.24  | -7.74  | -6.25  | -5.71  | -6.09  | 1.69   | -7.04  | -9.72  | -6.90  | 5.62   |                  |
| $\tilde{R}_{23}$     |                  |        | -3.64  | -3.77  | -6.65  | -4.87  | -7.61  | -3.39  | 5.20   | -4.48  | -10.86 | -3.00  | $\times 10^{-3}$ |
|                      |                  |        | -4.14  | -3.50  | -3.19  | 4.79   | -2.98  | -6.52  | -2.03  | 15.21  | -1.81  | -9.14  |                  |
|                      |                  |        | 1.64   | -4.90  | -8.18  | -5.87  | 12.63  | -4.53  | -11.31 | -5.49  | 18.27  | -4.14  |                  |
|                      |                  |        | -5.70  | 2.94   | -4.76  | -9.19  | -5.41  | 13.47  | -4.47  | -11.68 | -5.10  | 18.20  |                  |
|                      |                  |        | -9.00  | -5.48  | 1.97   | -5.87  | -12.01 | -5.25  | 12.74  | -5.63  | -13.78 | -5.01  |                  |
|                      |                  |        | -5.74  | -8.57  | -4.99  | 13.15  | -4.77  | -10.91 | -4.00  | 18.21  | -3.78  | -12.46 |                  |
|                      |                  |        | 10.12  | -6.53  | -12.40 | -6.79  | 17.58  | -6.28  | -14.04 | -6.54  | 17.46  | -6.02  |                  |
|                      |                  |        | -7.18  | 11.14  | -6.46  | -12.15 | -7.00  | 17.76  | -6.29  | -13.17 | -6.79  | 16.89  |                  |
|                      |                  |        | -12.85 | -7.07  | 10.27  | -6.80  | -14.26 | -6.95  | 17.46  | -6.68  | -14.07 | -6.81  |                  |
|                      |                  |        | -7.21  | -12.15 | -6.67  | 17.88  | -6.42  | -13.19 | -5.86  | 15.92  | -5.61  | -13.18 |                  |
| $\tilde{R}_{33}$     |                  |        | 9.40   | -7.97  | -14.43 | -7.63  | 18.53  | -7.81  | -14.08 | -7.48  | 11.84  | -7.62  | $\times 10^{-3}$ |
|                      |                  |        | -7.97  | 9.87   | -7.96  | -13.39 | -8.36  | 18.16  | -7.86  | -12.59 | -8.19  | 11.10  |                  |
|                      |                  |        | -14.43 | -7.96  | 9.40   | -7.65  | -13.93 | -8.41  | 18.36  | -7.62  | -11.41 | -8.31  |                  |
|                      |                  |        | -7.63  | -13.39 | -7.65  | 14.73  | -7.86  | -12.80 | -7.52  | 7.45   | -7.21  | -10.86 |                  |
|                      |                  |        | 18.53  | -8.36  | -13.93 | -7.86  | 16.57  | -9.02  | -11.10 | -8.28  | 0.65   | -8.85  |                  |
|                      |                  |        | -7.81  | 18.16  | -8.41  | -12.80 | -9.02  | 16.71  | -9.13  | -9.77  | -9.26  | 0.18   |                  |
|                      |                  |        | -14.08 | -7.86  | 18.36  | -7.52  | -11.10 | -9.13  | 16.46  | -8.43  | -5.51  | -9.42  |                  |
|                      |                  |        | -7.48  | -12.59 | -7.62  | 7.45   | -8.28  | -9.77  | -8.43  | 18.40  | -8.52  | -5.18  |                  |
|                      |                  |        | 11.84  | -8.19  | -11.41 | -7.21  | 0.65   | -9.26  | -5.51  | -8.52  | 18.61  | -9.65  |                  |
|                      |                  |        | -7.62  | 11.10  | -8.31  | -10.86 | -8.85  | 0.18   | -9.42  | -5.18  | -9.65  | 18.59  |                  |

TABLE V: The Coefficient Matrix  $\tilde{X}$  of Fault Detection in TABLE I

| Coefficient Matrices |                  |  | Values |       |       |       |       |       |       |       |       |       |
|----------------------|------------------|--|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| $\tilde{X}$          | $\tilde{X}_{11}$ |  | 5.50   | 10.98 | 10.94 | 10.92 | 10.89 | 10.86 | 10.84 | 10.82 | 10.80 | 10.77 |
|                      |                  |  | 10.98  | 5.48  | 10.92 | 10.89 | 10.86 | 10.84 | 10.81 | 10.79 | 10.76 | 10.74 |
|                      |                  |  | 10.94  | 10.92 | 5.44  | 10.86 | 10.84 | 10.81 | 10.78 | 10.75 | 10.73 | 10.71 |
|                      |                  |  | 10.92  | 10.89 | 10.86 | 5.42  | 10.80 | 10.78 | 10.77 | 10.73 | 10.70 | 10.68 |
|                      |                  |  | 10.89  | 10.86 | 10.84 | 10.80 | 5.39  | 10.75 | 10.73 | 10.70 | 10.67 | 10.65 |
|                      |                  |  | 10.86  | 10.84 | 10.81 | 10.78 | 10.75 | 5.36  | 10.70 | 10.67 | 10.65 | 10.63 |
|                      |                  |  | 10.84  | 10.81 | 10.78 | 10.77 | 10.73 | 10.70 | 5.34  | 10.65 | 10.62 | 10.60 |
|                      |                  |  | 10.82  | 10.79 | 10.75 | 10.73 | 10.70 | 10.67 | 10.65 | 5.31  | 10.60 | 10.57 |
|                      |                  |  | 10.80  | 10.76 | 10.73 | 10.70 | 10.67 | 10.65 | 10.62 | 10.60 | 5.29  | 10.55 |
|                      |                  |  | 10.77  | 10.74 | 10.71 | 10.68 | 10.65 | 10.63 | 10.60 | 10.57 | 10.55 | 5.26  |
|                      | $\tilde{X}_{12}$ |  | 10.75  | 10.72 | 10.71 | 10.69 | 10.67 | 10.65 | 10.64 | 10.62 | 10.60 | 10.58 |
|                      |                  |  | 10.72  | 10.69 | 10.68 | 10.65 | 10.64 | 10.62 | 10.61 | 10.58 | 10.57 | 10.56 |
|                      |                  |  | 10.68  | 10.67 | 10.65 | 10.63 | 10.61 | 10.59 | 10.57 | 10.56 | 10.54 | 10.52 |
|                      |                  |  | 10.66  | 10.63 | 10.62 | 10.59 | 10.57 | 10.56 | 10.54 | 10.53 | 10.51 | 10.49 |
|                      |                  |  | 10.63  | 10.61 | 10.59 | 10.57 | 10.55 | 10.53 | 10.51 | 10.49 | 10.47 | 10.46 |
|                      |                  |  | 10.60  | 10.58 | 10.57 | 10.54 | 10.52 | 10.50 | 10.48 | 10.47 | 10.45 | 10.43 |
|                      |                  |  | 10.58  | 10.55 | 10.54 | 10.52 | 10.49 | 10.47 | 10.45 | 10.44 | 10.42 | 10.40 |
|                      |                  |  | 10.55  | 10.53 | 10.51 | 10.48 | 10.46 | 10.44 | 10.43 | 10.40 | 10.39 | 10.38 |
|                      |                  |  | 10.52  | 10.50 | 10.48 | 10.46 | 10.44 | 10.42 | 10.40 | 10.38 | 10.37 | 10.35 |
|                      |                  |  | 10.50  | 10.47 | 10.46 | 10.43 | 10.41 | 10.39 | 10.37 | 10.35 | 10.34 | 10.33 |
|                      | $\tilde{X}_{13}$ |  | 10.57  | 10.56 | 10.55 | 10.54 | 10.53 | 10.51 | 10.51 | 10.50 | 10.49 | 10.49 |
|                      |                  |  | 10.54  | 10.52 | 10.51 | 10.50 | 10.49 | 10.49 | 10.47 | 10.47 | 10.46 | 10.46 |
|                      |                  |  | 10.51  | 10.49 | 10.49 | 10.47 | 10.46 | 10.45 | 10.44 | 10.43 | 10.43 | 10.43 |
|                      |                  |  | 10.48  | 10.46 | 10.45 | 10.44 | 10.43 | 10.42 | 10.41 | 10.41 | 10.39 | 10.39 |
|                      |                  |  | 10.45  | 10.43 | 10.42 | 10.41 | 10.40 | 10.39 | 10.38 | 10.37 | 10.36 | 10.36 |
|                      |                  |  | 10.42  | 10.41 | 10.39 | 10.38 | 10.37 | 10.36 | 10.35 | 10.33 | 10.33 | 10.33 |
|                      |                  |  | 10.39  | 10.37 | 10.36 | 10.35 | 10.34 | 10.33 | 10.32 | 10.31 | 10.31 | 10.30 |
|                      |                  |  | 10.36  | 10.35 | 10.33 | 10.32 | 10.31 | 10.29 | 10.29 | 10.28 | 10.28 | 10.27 |
|                      |                  |  | 10.33  | 10.32 | 10.31 | 10.30 | 10.28 | 10.27 | 10.26 | 10.26 | 10.26 | 10.24 |
|                      |                  |  | 10.31  | 10.30 | 10.27 | 10.27 | 10.26 | 10.25 | 10.23 | 10.23 | 10.22 | 10.22 |
|                      | $\tilde{X}_{22}$ |  | 5.24   | 10.45 | 10.43 | 10.41 | 10.39 | 10.37 | 10.34 | 10.33 | 10.31 | 10.30 |
|                      |                  |  | 10.45  | 5.21  | 10.41 | 10.38 | 10.37 | 10.34 | 10.33 | 10.30 | 10.30 | 10.28 |
|                      |                  |  | 10.43  | 10.41 | 5.19  | 10.36 | 10.34 | 10.32 | 10.31 | 10.28 | 10.27 | 10.25 |
|                      |                  |  | 10.41  | 10.38 | 10.36 | 5.17  | 10.32 | 10.29 | 10.28 | 10.26 | 10.25 | 10.23 |
|                      |                  |  | 10.39  | 10.37 | 10.34 | 10.32 | 5.15  | 10.28 | 10.26 | 10.24 | 10.22 | 10.20 |
|                      |                  |  | 10.37  | 10.34 | 10.32 | 10.29 | 10.28 | 5.13  | 10.24 | 10.21 | 10.20 | 10.19 |
|                      |                  |  | 10.34  | 10.33 | 10.31 | 10.28 | 10.26 | 10.24 | 5.11  | 10.20 | 10.19 | 10.17 |
|                      |                  |  | 10.33  | 10.30 | 10.28 | 10.26 | 10.24 | 10.21 | 10.20 | 5.09  | 10.17 | 10.15 |
|                      |                  |  | 10.31  | 10.30 | 10.27 | 10.25 | 10.22 | 10.20 | 10.19 | 10.17 | 5.08  | 10.13 |
|                      |                  |  | 10.30  | 10.28 | 10.25 | 10.23 | 10.20 | 10.19 | 10.17 | 10.15 | 10.13 | 5.06  |
|                      | $\tilde{X}_{23}$ |  | 10.28  | 10.26 | 10.26 | 10.25 | 10.23 | 10.22 | 10.21 | 10.21 | 10.20 | 10.19 |
|                      |                  |  | 10.26  | 10.25 | 10.23 | 10.22 | 10.21 | 10.20 | 10.19 | 10.18 | 10.16 | 10.17 |
|                      |                  |  | 10.24  | 10.22 | 10.21 | 10.20 | 10.19 | 10.17 | 10.17 | 10.15 | 10.15 | 10.14 |
|                      |                  |  | 10.21  | 10.20 | 10.19 | 10.17 | 10.16 | 10.15 | 10.14 | 10.13 | 10.12 | 10.12 |
|                      |                  |  | 10.19  | 10.17 | 10.16 | 10.15 | 10.14 | 10.13 | 10.11 | 10.11 | 10.10 | 10.08 |
|                      |                  |  | 10.17  | 10.16 | 10.14 | 10.13 | 10.12 | 10.10 | 10.09 | 10.08 | 10.08 | 10.07 |
|                      |                  |  | 10.15  | 10.14 | 10.12 | 10.11 | 10.10 | 10.09 | 10.08 | 10.06 | 10.06 | 10.05 |
|                      |                  |  | 10.13  | 10.12 | 10.10 | 10.09 | 10.08 | 10.06 | 10.06 | 10.05 | 10.04 | 10.03 |
|                      |                  |  | 10.11  | 10.09 | 10.09 | 10.08 | 10.07 | 10.05 | 10.04 | 10.03 | 10.02 | 10.01 |
|                      |                  |  | 10.09  | 10.08 | 10.07 | 10.06 | 10.04 | 10.03 | 10.02 | 10.01 | 10.01 | 9.99  |
|                      | $\tilde{X}_{33}$ |  | 5.04   | 10.06 | 10.05 | 10.04 | 10.02 | 10.01 | 10.00 | 9.99  | 9.99  | 9.98  |
|                      |                  |  | 10.06  | 5.03  | 10.03 | 10.02 | 10.02 | 10.00 | 9.99  | 9.98  | 9.97  | 9.96  |
|                      |                  |  | 10.05  | 10.03 | 5.01  | 10.01 | 9.99  | 9.98  | 9.98  | 9.96  | 9.96  | 9.95  |
|                      |                  |  | 10.04  | 10.02 | 10.01 | 5.00  | 9.98  | 9.97  | 9.96  | 9.95  | 9.94  | 9.93  |
|                      |                  |  | 10.02  | 10.02 | 9.99  | 9.98  | 4.98  | 9.96  | 9.95  | 9.94  | 9.93  | 9.92  |
|                      |                  |  | 10.01  | 10.00 | 9.98  | 9.97  | 9.96  | 4.98  | 9.93  | 9.92  | 9.92  | 9.91  |
|                      |                  |  | 10.00  | 9.99  | 9.98  | 9.96  | 9.95  | 9.93  | 4.96  | 9.92  | 9.91  | 9.90  |
|                      |                  |  | 9.99   | 9.98  | 9.96  | 9.95  | 9.94  | 9.92  | 9.92  | 4.95  | 9.89  | 9.90  |
|                      |                  |  | 9.99   | 9.97  | 9.96  | 9.94  | 9.93  | 9.92  | 9.91  | 9.89  | 4.95  | 9.89  |
|                      |                  |  | 9.98   | 9.96  | 9.95  | 9.93  | 9.92  | 9.91  | 9.90  | 9.90  | 9.89  | 4.94  |

TABLE VI: The Coefficient Matrix  $\tilde{Y}$  of Fault Detection in TABLE I

| Coefficient Matrices |  | Values |        |        |        |        |        |        |        |        |        |                  |  |  |  |  |
|----------------------|--|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|------------------|--|--|--|--|
| $\tilde{Y}_{11}$     |  | 32.36  | 4.73   | 14.21  | 116.21 | 33.77  | 6.12   | 15.61  | -21.89 | -18.26 | -0.81  | $\times 10^{-5}$ |  |  |  |  |
|                      |  | 32.63  | 4.52   | 14.63  | 116.95 | 34.04  | 5.92   | 16.03  | -22.21 | -18.40 | -1.08  |                  |  |  |  |  |
|                      |  | 32.89  | 4.31   | 15.04  | 117.69 | 34.31  | 5.71   | 16.45  | -22.54 | -18.55 | -1.36  |                  |  |  |  |  |
|                      |  | 33.16  | 4.11   | 15.46  | 118.43 | 34.58  | 5.50   | 16.87  | -22.87 | -18.69 | -1.63  |                  |  |  |  |  |
|                      |  | 33.43  | 3.90   | 15.88  | 119.17 | 34.85  | 5.30   | 17.29  | -23.19 | -18.84 | -1.90  |                  |  |  |  |  |
|                      |  | 33.70  | 3.69   | 16.30  | 119.91 | 35.12  | 5.09   | 17.71  | -23.52 | -18.99 | -2.17  |                  |  |  |  |  |
|                      |  | 33.97  | 3.49   | 16.72  | 120.65 | 35.39  | 4.88   | 18.13  | -23.84 | -19.13 | -2.43  |                  |  |  |  |  |
|                      |  | 34.24  | 3.28   | 17.14  | 121.40 | 35.66  | 4.68   | 18.55  | -24.16 | -19.28 | -2.70  |                  |  |  |  |  |
|                      |  | 34.51  | 3.08   | 17.56  | 122.15 | 35.93  | 4.47   | 18.96  | -24.48 | -19.42 | -2.97  |                  |  |  |  |  |
|                      |  | 34.79  | 2.87   | 17.97  | 122.90 | 36.20  | 4.27   | 19.38  | -24.80 | -19.57 | -3.24  |                  |  |  |  |  |
| $\tilde{Y}_{12}$     |  | -13.86 | -18.80 | -12.25 | 0.59   | -12.45 | -17.40 | -10.84 | 1.99   | -11.04 | -15.99 | $\times 10^{-5}$ |  |  |  |  |
|                      |  | -13.68 | -19.12 | -12.36 | 0.32   | -12.27 | -17.71 | -10.95 | 1.72   | -10.86 | -16.31 |                  |  |  |  |  |
|                      |  | -13.50 | -19.43 | -12.47 | 0.05   | -12.09 | -18.02 | -11.06 | 1.45   | -10.68 | -16.62 |                  |  |  |  |  |
|                      |  | -13.32 | -19.74 | -12.58 | -0.22  | -11.91 | -18.34 | -11.17 | 1.18   | -10.50 | -16.93 |                  |  |  |  |  |
|                      |  | -13.14 | -20.05 | -12.69 | -0.49  | -11.73 | -18.65 | -11.28 | 0.91   | -10.32 | -17.24 |                  |  |  |  |  |
|                      |  | -12.96 | -20.36 | -12.79 | -0.76  | -11.55 | -18.95 | -11.38 | 0.64   | -10.14 | -17.55 |                  |  |  |  |  |
|                      |  | -12.78 | -20.67 | -12.90 | -1.03  | -11.37 | -19.26 | -11.49 | 0.38   | -9.96  | -17.85 |                  |  |  |  |  |
|                      |  | -12.60 | -20.98 | -13.01 | -1.30  | -11.19 | -19.57 | -11.60 | 0.11   | -9.78  | -18.16 |                  |  |  |  |  |
|                      |  | -12.42 | -21.28 | -13.12 | -1.57  | -11.01 | -19.87 | -11.71 | -0.16  | -9.60  | -18.46 |                  |  |  |  |  |
|                      |  | -12.24 | -21.59 | -13.22 | -1.83  | -10.83 | -20.18 | -11.81 | -0.43  | -9.42  | -18.77 |                  |  |  |  |  |
| $\tilde{Y}_{13}$     |  | -9.44  | 3.39   | -9.64  | -14.59 | -8.03  | 4.79   | -8.23  | -13.19 | -6.62  | 6.19   | $\times 10^{-5}$ |  |  |  |  |
|                      |  | -9.54  | 3.12   | -9.46  | -14.90 | -8.14  | 4.52   | -8.05  | -13.50 | -6.73  | 5.92   |                  |  |  |  |  |
|                      |  | -9.65  | 2.85   | -9.28  | -15.21 | -8.24  | 4.25   | -7.87  | -13.81 | -6.84  | 5.65   |                  |  |  |  |  |
|                      |  | -9.76  | 2.58   | -9.10  | -15.53 | -8.35  | 3.98   | -7.69  | -14.12 | -6.94  | 5.38   |                  |  |  |  |  |
|                      |  | -9.87  | 2.31   | -8.92  | -15.83 | -8.46  | 3.71   | -7.51  | -14.43 | -7.05  | 5.12   |                  |  |  |  |  |
|                      |  | -9.97  | 2.05   | -8.74  | -16.14 | -8.57  | 3.45   | -7.33  | -14.73 | -7.16  | 4.85   |                  |  |  |  |  |
|                      |  | -10.08 | 1.78   | -8.56  | -16.45 | -8.67  | 3.18   | -7.15  | -15.04 | -7.26  | 4.58   |                  |  |  |  |  |
|                      |  | -10.19 | 1.51   | -8.38  | -16.75 | -8.78  | 2.91   | -6.97  | -15.34 | -7.37  | 4.31   |                  |  |  |  |  |
|                      |  | -10.29 | 1.24   | -8.20  | -17.06 | -8.89  | 2.64   | -6.79  | -15.65 | -7.48  | 4.05   |                  |  |  |  |  |
|                      |  | -10.40 | 0.98   | -8.02  | -17.36 | -8.99  | 2.38   | -6.61  | -15.95 | -7.58  | 3.78   |                  |  |  |  |  |
| $\tilde{Y}_{21}$     |  | 35.06  | 2.67   | 18.39  | 123.65 | 36.47  | 4.07   | 19.80  | -25.12 | -19.71 | -3.51  | $\times 10^{-5}$ |  |  |  |  |
|                      |  | 35.33  | 2.46   | 18.81  | 124.40 | 36.74  | 3.86   | 20.22  | -25.43 | -19.85 | -3.77  |                  |  |  |  |  |
|                      |  | 35.60  | 2.26   | 19.23  | 125.15 | 37.01  | 3.66   | 20.64  | -25.75 | -20.00 | -4.04  |                  |  |  |  |  |
|                      |  | 35.87  | 2.06   | 19.65  | 125.90 | 37.28  | 3.46   | 21.06  | -26.06 | -20.14 | -4.30  |                  |  |  |  |  |
|                      |  | 36.14  | 1.86   | 20.07  | 126.66 | 37.56  | 3.25   | 21.48  | -26.37 | -20.29 | -4.57  |                  |  |  |  |  |
|                      |  | 36.41  | 1.65   | 20.49  | 127.42 | 37.83  | 3.05   | 21.90  | -26.68 | -20.43 | -4.83  |                  |  |  |  |  |
|                      |  | 36.68  | 1.45   | 20.91  | 128.18 | 38.10  | 2.85   | 22.32  | -26.99 | -20.57 | -5.10  |                  |  |  |  |  |
|                      |  | 36.96  | 1.25   | 21.33  | 128.94 | 38.37  | 2.65   | 22.74  | -27.30 | -20.72 | -5.36  |                  |  |  |  |  |
|                      |  | 37.23  | 1.05   | 21.75  | 129.70 | 38.65  | 2.45   | 23.16  | -27.60 | -20.86 | -5.62  |                  |  |  |  |  |
|                      |  | 37.50  | 0.85   | 22.16  | 130.47 | 38.92  | 2.25   | 23.58  | -27.90 | -21.00 | -5.89  |                  |  |  |  |  |
| $\tilde{Y}_{22}$     |  | -12.06 | -21.89 | -13.33 | -2.10  | -10.65 | -20.48 | -11.92 | -0.69  | -9.24  | -19.07 | $\times 10^{-5}$ |  |  |  |  |
|                      |  | -11.88 | -22.19 | -13.44 | -2.37  | -10.47 | -20.78 | -12.03 | -0.96  | -9.07  | -19.37 |                  |  |  |  |  |
|                      |  | -11.70 | -22.49 | -13.55 | -2.63  | -10.29 | -21.08 | -12.13 | -1.23  | -8.89  | -19.67 |                  |  |  |  |  |
|                      |  | -11.52 | -22.79 | -13.65 | -2.90  | -10.11 | -21.38 | -12.24 | -1.49  | -8.71  | -19.97 |                  |  |  |  |  |
|                      |  | -11.34 | -23.09 | -13.76 | -3.16  | -9.93  | -21.67 | -12.34 | -1.76  | -8.53  | -20.26 |                  |  |  |  |  |
|                      |  | -11.16 | -23.38 | -13.86 | -3.43  | -9.76  | -21.97 | -12.45 | -2.02  | -8.35  | -20.55 |                  |  |  |  |  |
|                      |  | -10.98 | -23.68 | -13.97 | -3.69  | -9.58  | -22.26 | -12.55 | -2.28  | -8.17  | -20.84 |                  |  |  |  |  |
|                      |  | -10.81 | -23.97 | -14.08 | -3.95  | -9.40  | -22.54 | -12.66 | -2.55  | -7.99  | -21.14 |                  |  |  |  |  |
|                      |  | -10.63 | -24.26 | -14.18 | -4.22  | -9.22  | -22.84 | -12.76 | -2.81  | -7.81  | -21.43 |                  |  |  |  |  |
|                      |  | -10.45 | -24.55 | -14.28 | -4.48  | -9.04  | -23.13 | -12.87 | -3.07  | -7.63  | -21.71 |                  |  |  |  |  |
| $\tilde{Y}_{23}$     |  | -10.51 | 0.71   | -7.84  | -17.66 | -9.10  | 2.11   | -6.43  | -16.25 | -7.69  | 3.52   | $\times 10^{-5}$ |  |  |  |  |
|                      |  | -10.61 | 0.44   | -7.66  | -17.96 | -9.20  | 1.85   | -6.25  | -16.55 | -7.79  | 3.25   |                  |  |  |  |  |
|                      |  | -10.72 | 0.18   | -7.48  | -18.26 | -9.31  | 1.58   | -6.07  | -16.85 | -7.90  | 2.99   |                  |  |  |  |  |
|                      |  | -10.83 | -0.09  | -7.30  | -18.55 | -9.41  | 1.32   | -5.89  | -17.14 | -8.00  | 2.72   |                  |  |  |  |  |
|                      |  | -10.93 | -0.35  | -7.12  | -18.85 | -9.52  | 1.05   | -5.72  | -17.44 | -8.11  | 2.46   |                  |  |  |  |  |
|                      |  | -11.04 | -0.61  | -6.94  | -19.14 | -9.62  | 0.79   | -5.54  | -17.73 | -8.21  | 2.20   |                  |  |  |  |  |
|                      |  | -11.14 | -0.88  | -6.76  | -19.43 | -9.73  | 0.53   | -5.36  | -18.02 | -8.31  | 1.93   |                  |  |  |  |  |
|                      |  | -11.25 | -1.14  | -6.58  | -19.72 | -9.83  | 0.26   | -5.18  | -18.31 | -8.42  | 1.67   |                  |  |  |  |  |
|                      |  | -11.35 | -1.40  | -6.41  | -20.01 | -9.93  | 0.00   | -5.00  | -18.60 | -8.52  | 1.41   |                  |  |  |  |  |
|                      |  | -11.45 | -1.67  | -6.23  | -20.30 | -10.04 | -0.26  | -4.82  | -18.88 | -8.62  | 1.15   |                  |  |  |  |  |
| $\tilde{Y}_{31}$     |  | 37.77  | 0.65   | 22.58  | 131.24 | 39.19  | 2.05   | 24.00  | -28.20 | -21.14 | -6.15  | $\times 10^{-5}$ |  |  |  |  |
|                      |  | 38.05  | 0.45   | 23.00  | 132.01 | 39.47  | 1.85   | 24.42  | -28.50 | -21.28 | -6.41  |                  |  |  |  |  |
|                      |  | 38.32  | 0.24   | 23.42  | 132.78 | 39.74  | 1.65   | 24.83  | -28.80 | -21.43 | -6.67  |                  |  |  |  |  |
|                      |  | 38.60  | 0.04   | 23.83  | 133.55 | 40.02  | 1.45   | 25.25  | -29.10 | -21.57 | -6.93  |                  |  |  |  |  |
|                      |  | 38.87  | -0.15  | 24.25  | 134.33 | 40.29  | 1.25   | 25.67  | -29.39 | -21.71 | -7.19  |                  |  |  |  |  |
|                      |  | 39.15  | -0.35  | 24.67  | 135.10 | 40.57  | 1.05   | 26.09  | -29.68 | -21.85 | -7.45  |                  |  |  |  |  |
|                      |  | 39.42  | -0.55  | 25.08  | 135.88 | 40.84  | 0.86   | 26.50  | -29.97 | -21.99 | -7.71  |                  |  |  |  |  |
|                      |  | 39.70  | -0.75  | 25.50  | 136.66 | 41.12  | 0.66   | 26.92  | -30.26 | -22.13 | -7.97  |                  |  |  |  |  |
|                      |  | 39.97  | -0.95  | 25.91  | 137.45 | 41.40  | 0.46   | 27.34  | -30.54 | -22.26 | -8.23  |                  |  |  |  |  |
|                      |  | 40.25  | -1.15  | 26.33  | 138.23 | 41.67  | 0.27   | 27.75  | -30.82 | -22.40 | -8.49  |                  |  |  |  |  |
| $\tilde{Y}_{32}$     |  | -10.27 | -24.84 | -14.39 | -4.74  | -8.87  | -23.42 | -12.97 | -3.33  | -7.46  | -22.00 | $\times 10^{-5}$ |  |  |  |  |
|                      |  | -10.10 | -25.12 | -14.49 | -5.00  | -8.69  | -23.70 | -13.08 | -3.59  | -7.28  | -22.28 |                  |  |  |  |  |
|                      |  | -9.92  | -25.40 | -14.60 | -5.26  | -8.51  | -23.98 | -13.18 | -3.85  | -7.10  | -22.56 |                  |  |  |  |  |
|                      |  | -9.74  | -25.68 | -14.70 | -5.52  | -8.34  | -24.26 | -13.28 | -4.12  | -6.93  | -22.84 |                  |  |  |  |  |
|                      |  | -9.57  | -25.96 | -14.80 | -5.78  | -8.16  | -24.54 | -13.38 | -4.38  | -6.75  | -23.12 |                  |  |  |  |  |
|                      |  | -9.39  | -26.24 | -14.91 | -6.04  | -7.99  | -24.82 | -13.49 | -4.64  | -6.58  | -23.40 |                  |  |  |  |  |
|                      |  | -9.22  | -26.52 | -15.01 | -6.30  | -7.81  | -25.09 | -13.59 | -4.89  | -6.40  | -23.67 |                  |  |  |  |  |
|                      |  | -9.05  | -26.79 | -15.11 | -6.56  | -7.64  | -25.36 | -13.69 | -5.15  | -6.23  | -23.94 |                  |  |  |  |  |
|                      |  | -8.87  | -27.06 | -15.21 | -6.82  | -7.46  | -25.63 | -13.79 | -5.41  | -6.05  | -24.21 |                  |  |  |  |  |
|                      |  | -8.70  | -27.33 | -15.31 | -7.08  | -7.29  | -25.90 | -13.89 | -5.67  | -5.88  | -24.48 |                  |  |  |  |  |
| $\tilde{Y}_{33}$     |  | -11.56 | -1.93  | -6.05  | -20.58 | -10.14 | -0.52  | -4.64  | -19.17 | -8.73  | 0.89   | $\times 10^{-5}$ |  |  |  |  |
|                      |  | -11.66 | -2.19  | -5.87  | -20.86 | -10.24 | -0.78  | -4.47  | -19.45 | -8.83  | 0.63   |                  |  |  |  |  |
|                      |  | -11.76 | -2.45  | -5.70  | -21.14 | -10.35 | -1.04  | -4.29  | -19.73 | -8.93  | 0.37   |                  |  |  |  |  |
|                      |  | -11.86 | -2.71  | -5.52  | -21.42 | -10.45 | -1.30  | -4.11  | -20.05 | -9.03  | 0.11   |                  |  |  |  |  |
|                      |  | -11.97 | -2.97  | -5.34  | -21.70 | -10.55 | -1.56  | -3.94  | -20.28 | -9.13  | -0.15  |                  |  |  |  |  |
|                      |  | -12.07 | -3.23  | -5.17  | -21.97 | -10.65 | -1.82  | -3.76  | -20.55 | -9.23  | -0.41  |                  |  |  |  |  |
|                      |  | -12.17 | -3.49  | -4.99  | -22.25 | -10.75 | -2.08  | -3.59  | -20.83 | -9.33  | -0.67  |                  |  |  |  |  |
|                      |  | -12.27 | -3.74  | -4.82  | -22.52 | -10.85 | -2.34  | -3.41  | -21.10 | -9.43  | -0.93  |                  |  |  |  |  |
|                      |  | -12.37 | -4.00  | -4.65  | -22.79 | -10.95 | -2.60  | -3.24  | -21.36 | -9.53  | -1.19  |                  |  |  |  |  |
|                      |  | -12.47 | -4.26  | -4.47  | -23.05 | -11.05 | -2.85  | -3.07  | -21.63 | -9.63  | -1.45  |                  |  |  |  |  |

TABLE VII: The Coefficient Matrix  $\tilde{Z}$  of Fault Detection in TABLE I

| Coefficient Matrices |                  |  | Values  |        |         |         |         |         |         |         |         |         |                  |
|----------------------|------------------|--|---------|--------|---------|---------|---------|---------|---------|---------|---------|---------|------------------|
| $\tilde{Z}$          | $Z_{11}$         |  | 25.79   | 11.07  | 54.01   | 104.64  | 52.06   | 11.71   | 55.07   | 24.57   | 12.75   | 1.95    | $\times 10^{-4}$ |
|                      |                  |  | 11.07   | -19.84 | 23.76   | 34.55   | 11.88   | -38.92  | 24.48   | -7.81   | -5.62   | -14.19  |                  |
|                      |                  |  | 54.01   | 23.76  | 30.92   | 95.21   | 46.73   | 16.30   | 55.75   | 25.11   | 14.90   | 8.44    |                  |
|                      |                  |  | 104.64  | 34.55  | 95.21   | 157.61  | 110.91  | 41.73   | 101.56  | 119.49  | 34.54   | 9.36    |                  |
|                      |                  |  | 52.06   | 11.88  | 46.73   | 110.91  | 26.58   | 12.51   | 47.75   | 31.28   | 13.31   | 2.87    |                  |
|                      |                  |  | 11.71   | -38.92 | 16.30   | 41.73   | 12.51   | -19.08  | 17.07   | -1.09   | -5.07   | -13.36  |                  |
|                      |                  |  | 55.07   | 24.48  | 55.75   | 101.56  | 47.75   | 17.07   | 23.57   | 32.15   | 15.57   | 9.29    |                  |
|                      |                  |  | 24.57   | -7.81  | 25.11   | 119.49  | 31.28   | -1.09   | 32.15   | -44.22  | -43.56  | -28.76  |                  |
|                      |                  |  | 12.75   | -5.62  | 14.90   | 34.54   | 13.31   | -5.07   | 15.57   | -43.56  | -11.09  | -12.79  |                  |
|                      |                  |  | 1.95    | -14.19 | 8.44    | 9.36    | 2.87    | -13.36  | 9.29    | -28.76  | -12.79  | 4.24    |                  |
|                      | $\tilde{Z}_{12}$ |  | 33.89   | 24.17  | 15.50   | 2.50    | 34.83   | 24.85   | 16.13   | 3.07    | 35.77   | 25.56   | $\times 10^{-4}$ |
|                      |                  |  | 24.80   | -2.11  | 3.79    | -13.36  | 25.51   | -1.18   | 4.73    | -12.53  | 26.20   | -0.24   |                  |
|                      |                  |  | 34.49   | 19.78  | 11.16   | 1.29    | 27.03   | 13.21   | 4.81    | -5.32   | 19.67   | 7.23    |                  |
|                      |                  |  | 48.15   | 125.47 | 42.90   | 16.43   | 55.34   | 131.01  | 49.88   | 23.67   | 62.46   | 136.23  |                  |
|                      |                  |  | 26.55   | 30.88  | 16.13   | 3.40    | 27.41   | 31.69   | 16.80   | 3.97    | 28.29   | 32.52   |                  |
|                      |                  |  | 17.37   | 4.47   | 4.22    | -12.53  | 18.14   | 5.44    | 5.14    | -11.69  | 18.89   | 6.41    |                  |
|                      |                  |  | 27.03   | 26.62  | 11.75   | 2.22    | 19.67   | 19.77   | 5.23    | -4.40   | 12.49   | 13.22   |                  |
|                      |                  |  | -18.26  | -81.02 | -34.68  | -21.86  | -11.55  | -74.19  | -27.44  | -15.17  | -5.22   | -67.14  |                  |
|                      |                  |  | -2.86   | -43.80 | -19.77  | -12.18  | -2.46   | -42.81  | -19.02  | -11.60  | -2.01   | -41.83  |                  |
|                      |                  |  | 9.61    | -23.70 | -4.25   | 9.77    | 10.46   | -22.99  | -3.34   | 10.63   | 11.29   | -22.27  |                  |
|                      | $\tilde{Z}_{13}$ |  | 16.80   | 3.68   | 36.72   | 26.29   | 17.49   | 4.32    | 37.67   | 27.05   | 18.21   | 4.98    | $\times 10^{-4}$ |
|                      |                  |  | 5.68    | -11.69 | 26.89   | 0.70    | 6.64    | -10.85  | 27.57   | 1.67    | 7.59    | -10.00  |                  |
|                      |                  |  | -1.89   | -11.81 | 12.49   | 0.27    | -9.15   | -18.64  | 5.58    | -7.07   | -16.61  | -25.72  |                  |
|                      |                  |  | 56.89   | 31.02  | 69.44   | 141.11  | 63.89   | 38.39   | 76.23   | 145.59  | 70.80   | 45.73   |                  |
|                      |                  |  | 17.49   | 4.57   | 29.17   | 33.36   | 18.21   | 5.20    | 30.07   | 34.22   | 18.95   | 5.85    |                  |
|                      |                  |  | 6.06    | -10.85 | 19.64   | 7.38    | 6.99    | -10.00  | 20.37   | 8.35    | 7.93    | -9.14   |                  |
|                      |                  |  | -1.34   | -10.98 | 5.58    | 7.21    | -8.46   | -17.88  | -1.00   | 0.26    | -15.82  | -25.02  |                  |
|                      |                  |  | -20.33  | -8.81  | 1.64    | -59.91  | -13.40  | -2.37   | 8.76    | -52.55  | -6.85   | 4.83    |                  |
|                      |                  |  | -18.29  | -11.06 | -1.52   | -40.85  | -17.59  | -10.55  | -0.98   | -39.88  | -16.91  | -10.07  |                  |
|                      |                  |  | -2.43   | 11.47  | 12.12   | -21.54  | -1.51   | 12.30   | 12.93   | -20.80  | -0.58   | 13.13   |                  |
|                      | $\tilde{Z}_{22}$ |  | 3.42    | -23.86 | -6.81   | 2.53    | 0.14    | -31.04  | -13.70  | -4.08   | -6.35   | -38.36  | $\times 10^{-4}$ |
|                      |                  |  | -23.86  | -36.60 | -35.06  | -16.81  | -16.90  | -66.92  | -27.81  | -10.17  | -10.19  | -59.68  |                  |
|                      |                  |  | -6.81   | -35.06 | -8.31   | -3.77   | -6.31   | -34.15  | -16.22  | -3.35   | -5.83   | -33.25  |                  |
|                      |                  |  | 2.53    | -16.81 | -3.77   | 5.15    | 3.45    | -16.04  | -2.85   | 11.47   | 4.36    | -15.25  |                  |
|                      |                  |  | 0.14    | -16.90 | -6.31   | 3.45    | -3.19   | -23.89  | -13.04  | -3.15   | -13.05  | -31.10  |                  |
|                      |                  |  | -31.04  | -66.92 | -34.15  | -16.04  | -23.89  | -29.22  | -26.99  | -9.31   | -16.91  | -52.30  |                  |
|                      |                  |  | -13.70  | -27.81 | -16.22  | -2.85   | -13.04  | -26.99  | -7.75   | -2.41   | -12.42  | -26.18  |                  |
|                      |                  |  | -4.08   | -10.17 | -3.35   | 11.47   | -3.15   | -9.31   | -2.41   | 6.05    | -2.22   | -8.44   |                  |
|                      |                  |  | -6.35   | -10.19 | -5.83   | 4.36    | -13.05  | -16.91  | -12.42  | -2.22   | -9.87   | -23.92  |                  |
|                      |                  |  | -38.36  | -59.68 | -33.25  | -15.25  | -31.10  | -52.30  | -26.18  | -8.44   | -23.92  | -21.92  |                  |
|                      | $\tilde{Z}_{23}$ |  | -20.85  | -10.64 | -13.05  | -45.76  | -28.19  | -17.53  | -20.05  | -53.16  | -35.65  | -24.66  | $\times 10^{-4}$ |
|                      |                  |  | -20.69  | -3.92  | -3.86   | -52.30  | -13.77  | 3.06    | 3.02    | -44.85  | -7.38   | 10.23   |                  |
|                      |                  |  | -15.55  | -2.97  | -5.40   | -32.36  | -14.90  | -2.61   | -5.02   | -31.49  | -14.29  | -2.11   |                  |
|                      |                  |  | -1.93   | 12.30  | 5.27    | -14.45  | -1.00   | 13.13   | 6.17    | -13.65  | -0.06   | 13.95   |                  |
|                      |                  |  | -20.08  | -9.81  | -20.05  | -38.44  | -27.32  | -16.77  | -27.26  | -45.85  | -34.71  | -23.96  |                  |
|                      |                  |  | -19.98  | -2.93  | -10.18  | -44.85  | -13.21  | 3.96    | -3.86   | -37.38  | -7.02   | 11.07   |                  |
|                      |                  |  | -14.90  | -2.01  | -11.82  | -25.40  | -14.29  | -1.62   | -11.26  | -24.64  | -13.71  | -1.14   |                  |
|                      |                  |  | -1.47   | 13.13  | -1.29   | -7.57   | -0.52   | 13.95   | -0.34   | -6.69   | 0.43    | 14.76   |                  |
|                      |                  |  | -19.33  | -8.96  | -27.26  | -31.14  | -26.47  | -15.99  | -34.61  | -38.50  | -33.78  | -23.24  |                  |
|                      |                  |  | -19.30  | -1.95  | -16.91  | -37.38  | -12.67  | 4.86    | -10.15  | -29.95  | -6.69   | 11.90   |                  |
|                      | $\tilde{Z}_{33}$ |  | -72.30  | -10.46 | -185.98 | -186.46 | -137.08 | -6.47   | -178.96 | -180.23 | -131.61 | -1.71   | $\times 10^{-4}$ |
|                      |                  |  | -10.46  | 69.38  | -81.07  | -9.80   | -0.89   | 147.57  | -72.46  | -0.13   | 8.72    | 155.61  |                  |
|                      |                  |  | -185.98 | -81.07 | -168.67 | -239.45 | -256.35 | -152.12 | -420.26 | -311.86 | -328.58 | -225.19 |                  |
|                      |                  |  | -186.46 | -9.80  | -239.45 | -147.63 | -121.96 | 57.42   | -169.12 | -226.29 | -62.46  | 127.16  |                  |
|                      |                  |  | -137.08 | -0.89  | -256.35 | -121.96 | -67.43  | 3.19    | -248.18 | -117.69 | -126.51 | 7.81    |                  |
|                      |                  |  | -6.47   | 147.57 | -152.12 | 57.42   | 3.19    | 78.23   | -144.18 | 66.21   | 12.81   | 163.59  |                  |
|                      |                  |  | -178.96 | -72.46 | -420.26 | -169.12 | -248.18 | -144.18 | -240.86 | -239.62 | -319.50 | -217.81 |                  |
|                      |                  |  | -180.23 | -0.13  | -311.86 | -226.29 | -117.69 | 66.21   | -239.62 | -78.60  | -57.61  | 135.23  |                  |
|                      |                  |  | -131.61 | 8.72   | -328.58 | -62.46  | -126.51 | 12.81   | -319.50 | -57.61  | -62.95  | 17.26   |                  |
|                      |                  |  | -1.71   | 155.61 | -225.19 | 127.16  | 7.81    | 163.59  | -217.81 | 135.23  | 17.26   | 87.02   |                  |